

ECE 3337 — Cumulative Practice Final

Signals & Systems — covers Midterm 1 topics (LTI systems, convolution, Fourier series/transform, BIBO stability) and Midterm 2 topics (sampling, reconstruction, filtering, Laplace transforms), following the professor's exam style.

1. (5 points)

What is the equivalent impulse response of the cascade of systems shown below?

$$\delta(t + 1) \longrightarrow te^{-3t}u(t) \longrightarrow \frac{d}{dt} \longrightarrow$$

2. (5 points)

Is the system $y(t) = tx(t - 1)$ linear time invariant? Explain your reasons.

3. (10 points)

Starting with the original sine/cosine Fourier representation, derive the exponential Fourier series and its coefficient formula. Then use your result to compute Parseval's theorem for the exponential form.

4. (5 points)

Compute the Fourier transform of $x(t) = e^{-2|t|} \sin(\omega_0 t)$. Use the properties of the Fourier transform.

5. (5 points)

The spectrum of a bird call signal is negligible above 9 kHz. What is the Nyquist sampling rate required to reconstruct the signal from its samples? Now suppose it's instead sampled at 15 kHz. Sketch the spectrum of the sampled signal and state whether reconstruction is possible.

6. (5 points)

Signal $x(t) = \sin(20\pi t) + \sin(50\pi t)$ is sampled at 40 samples/s. The result is passed through an ideal brick-wall lowpass filter with a cutoff frequency of 20 Hz. Compute and sketch the spectrum of the output signal.

7. (10 points)

A commonly used method for converting discrete-time signals to continuous-time signals holds $x(t)$ constant between samples taken at $t = nT_s$.

(a) Compute the frequency response $\mathbf{H}(\omega)$ of this interpolation method.

(b) This method is used to reconstruct $\sin(0.2t/T_s)$ from samples at $t = nT_s$. The first copy of the spectrum induced by sampling lies in $\frac{2\pi - 0.2}{T_s} \leq \omega \leq \frac{2\pi + 0.2}{T_s}$. Compute the reduction in gain between $\omega = 0$ and the lower edge of that copy.

8. (5 points)

Find $X(s)$, the Laplace transform of the signal

$$x(t) = 3\delta(2t - 6) + t^2 u(t - 2) + te^{2-3t} u(t)$$

(Reference: use the Laplace transform tables from your course packet — Table of transform pairs and Table of properties.)

9. (5 points)

Show that

$$\mathcal{L} \left[\int_{0^-}^t x(\tau) d\tau \right] = \frac{1}{s} X(s)$$

10. (5 points)

How do you define a bounded signal? Derive the condition an LTI system must satisfy to be BIBO stable, and state — without re-deriving — how this relates to the poles of a system's Laplace transfer function $H(s)$.

ANSWER GUIDE

For each problem: **what to recognize** (the pattern-matching cue that tells you which tool to reach for), then the **full worked solution**.

1. Cascade impulse response

Recognize: three blocks in series $\rightarrow$ convolve all impulse responses together, in any order (convolution is associative and commutative). One block is a differentiator, which is a shortcut: convolving *anything* with d/dt 's impulse response is the same as just differentiating that signal directly. And convolving with a plain delta $\delta(t - a)$ is a pure shift — never integrate for that step.

Step 1 — combine the shift and the exponential term. $\delta(t + 1) * te^{-3t}u(t)$: convolving with $\delta(t + 1)$ shifts the second signal left by 1:

$$\delta(t + 1) * te^{-3t}u(t) = (t + 1)e^{-3(t+1)}u(t + 1)$$

Step 2 — apply the differentiator. The last block means: differentiate the result of Step 1.

$$\frac{d}{dt} \left[(t + 1)e^{-3(t+1)}u(t + 1) \right]$$

Use the product rule on $(t + 1)e^{-3(t+1)}$, remembering $u(t + 1)$ itself also needs the product rule (its derivative is $\delta(t + 1)$):

$$\frac{d}{dt} \left[(t + 1)e^{-3(t+1)} \right] u(t + 1) + (t + 1)e^{-3(t+1)} \cdot \delta(t + 1)$$

For the first piece, product rule on $(t + 1)$ and $e^{-3(t+1)}$:

$$e^{-3(t+1)} + (t + 1)(-3)e^{-3(t+1)} = e^{-3(t+1)} [1 - 3(t + 1)]$$

For the second piece, sifting: $(t + 1)e^{-3(t+1)}\delta(t + 1)$ evaluates the non-delta factor at $t = -1$: $(-1 + 1)e^{-3(0)} = 0$. This term vanishes.

$$h(t) = [1 - 3(t + 1)]e^{-3(t+1)}u(t + 1)$$

2. Linearity and time-invariance of $y(t) = t x(t - 1)$

Recognize: any system where the output is multiplied by an explicit function of t (not a constant) is immediately suspicious for time-invariance — a "coefficient" that depends on absolute time almost always breaks TI. Linearity, by contrast, survives multiplication by t because t doesn't care whether you scale/add signals first or afterward.

Linearity check: for inputs x_1, x_2 and scalars α, β :

$$T\{\alpha x_1 + \beta x_2\} = t[\alpha x_1(t - 1) + \beta x_2(t - 1)] = \alpha [t x_1(t - 1)] + \beta [t x_2(t - 1)] = \alpha T\{x_1\} + \beta T\{x_2\}$$

✓ **Linear.**

Time-invariance check: let $x_2(t) = x(t - t_0)$. Then

$$T\{x_2\}(t) = t x_2(t - 1) = t x(t - 1 - t_0)$$

But the true shifted output is

$$y(t - t_0) = (t - t_0) x(t - t_0 - 1)$$

Compare the leading coefficients: t vs. $(t - t_0)$ — not equal unless $t_0 = 0$. **✗ Not time-invariant.**

$$\text{Linear, not time-invariant}$$

3. Exponential Fourier series derivation + Parseval

Recognize: "derive the exponential form" is always a two-part answer: (1) rewrite cos, sin using Euler's formula and regroup by exponential, (2) derive the coefficient formula separately using the **orthogonality integral** — multiply both sides by $e^{-jm\omega_0 t}$, integrate over one period, and note that only the $n = m$ term survives.

(Full mechanical steps for both parts were worked in your midterm review — same procedure applies here verbatim; if you want it fully re-derived line by line again, ask and I'll redo it in full.)

****For Parseval****, once you have x_n , the recognize-cue is: "average power over one period" → write $P = \frac{1}{T_0} \int |x(t)|^2 dt = \frac{1}{T_0} \int x(t)x^*(t)dt$, substitute the *conjugated* series for $x^*(t)$, swap sum/integral, and notice the leftover integral is literally the definition of x_n — giving:

$$\boxed{\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |x_n|^2}$$

4. FT of $x(t) = e^{-2|t|} \sin(\omega_0 t)$

Recognize: $e^{-a|t|}$ is a *known even* transform pair — don't integrate it from scratch, look it up (or rederive quickly via splitting into $u(t)$ and $u(-t)$ pieces). Then $\times \sin(\omega_0 t)$ is a **modulation** — this is a strong recognize-cue: multiplying by sin or cos of a fixed frequency always means "shift the spectrum by $\pm\omega_0$ and combine," never re-integrate the whole product directly.

Step 1 — base transform:

$$\mathcal{F}\{e^{-2|t|}\} = \frac{2(2)}{4 + \omega^2} = \frac{4}{4 + \omega^2}$$

(general form $\mathcal{F}\{e^{-a|t|}\} = \frac{2a}{a^2 + \omega^2}$, here $a = 2$)

Step 2 — modulation property for sin:

$$\mathcal{F}\{x(t) \sin(\omega_0 t)\} = \frac{1}{2j} [X(\omega - \omega_0) - X(\omega + \omega_0)]$$

Step 3 — substitute:

$$X(\omega) = \frac{1}{2j} \left[\frac{4}{4 + (\omega - \omega_0)^2} - \frac{4}{4 + (\omega + \omega_0)^2} \right]$$

$$\boxed{X(\omega) = \frac{2}{j} \left[\frac{1}{4 + (\omega - \omega_0)^2} - \frac{1}{4 + (\omega + \omega_0)^2} \right]}$$

5. Nyquist rate + sketch at 15 kHz

Recognize: "negligible above X Hz" always means the signal's highest frequency is X — that's your f_0 for Nyquist purposes. "Sketch and state if reconstruction is possible" is always the same 3-part check: draw replicas at $k f_s \pm f_0$, mark the Nyquist band $\pm f_s/2$, see if anything besides the true spike falls inside it.

Nyquist rate: $f_0 = 9 \text{ kHz} \Rightarrow f_{\text{Nyquist}} = 2f_0 = \boxed{18 \text{ kHz}}$

At $f_s = 15 \text{ kHz}$: since $15 < 18$, this is **below** Nyquist. Replicas land at $15k \pm 9$: $k = 0 \rightarrow \pm 9$; $k = 1 \rightarrow 6, 24$; $k = -1 \rightarrow -6, -24$. The $k = 1$ replica at 6 kHz falls **inside** the Nyquist band ($-7.5, 7.5$) kHz — aliasing occurs. **Not reconstructible**; the recovered signal would show energy at 6 kHz that isn't really there.

6. Sampling + brick-wall filter output spectrum

Recognize: this is the exact same replica-and-filter mechanic as the Nyquist checks above, just carried one step further — after finding all replica locations, you literally **erase** anything outside the filter's cutoff, and whatever's left inside the cutoff is the output spectrum.

Original frequencies: 10 Hz and 25 Hz (from $\sin(20\pi t) = \sin(2\pi \cdot 10t)$ and $\sin(2\pi \cdot 25t)$). Sampled at $f_s = 40$: replicas at $40k \pm 10$ and $40k \pm 25$.

- ± 10 : $k = 0 \rightarrow$ inside ± 20 Hz cutoff. **Kept**.
- ± 25 : $k = 0 \rightarrow 25 > 20$, outside cutoff. **Removed** directly.
- $k = 1$ replicas of ± 10 : 30, 50 $\rightarrow$ outside cutoff, ignore.
- $k = -1$ replica of ± 25 : $40 - 25 = 15 \rightarrow$ **inside** the ± 20 Hz cutoff! This is aliasing landing right in the passband.

Output spectrum: spikes at ± 10 Hz (the true low-frequency content) **and** ± 15 Hz (an aliased copy of the 25 Hz tone folded down by sampling). The 25 Hz component itself is gone, but it left behind a "ghost" at 15 Hz that the filter can't distinguish from a real signal.

Output = $\sin(20\pi t) + \sin(30\pi t)$ -equivalent alias at 15 Hz (with a sign/phase factor from folding)

7. Zero-order hold frequency response + gain reduction

Recognize: "held constant between samples" is always the **zero-order hold** — its impulse response is a single rectangular pulse of width T_s , so its frequency response is always a **sinc function** (Fourier transform of a rectangle). "Compute the reduction in gain between two frequencies" always means: evaluate $|H(\omega)|$ at both points and take the ratio.

Part (a): ZOH impulse response: $h(t) = u(t) - u(t - T_s)$ — a rectangular pulse of height 1, width T_s .

$$H(\omega) = \int_0^{T_s} e^{-j\omega t} dt = \frac{1 - e^{-j\omega T_s}}{j\omega}$$

Factor out to reveal a sinc pattern (multiply/divide by $e^{-j\omega T_s/2}$):

$$H(\omega) = T_s e^{-j\omega T_s/2} \frac{\sin(\omega T_s/2)}{\omega T_s/2} = T_s e^{-j\omega T_s/2} \text{sinc}\left(\frac{\omega T_s}{2\pi}\right)$$

$$H(\omega) = T_s e^{-j\omega T_s/2} \text{sinc}\left(\frac{\omega T_s}{2\pi}\right)$$

Part (b): magnitude only matters for gain (the exponential phase term has magnitude 1, so ignore it):

$$|H(\omega)| = T_s \left| \text{sinc}\left(\frac{\omega T_s}{2\pi}\right) \right|$$

Gain ratio = $\frac{|H(\omega_1)|}{|H(0)|}$ where $\omega_1 = \frac{2\pi - 0.2}{T_s}$ and $H(0) = T_s$ ($\text{sinc}(0) = 1$):

$$\frac{|H(\omega_1)|}{|H(0)|} = \left| \text{sinc}\left(\frac{(2\pi - 0.2)/T_s \cdot T_s}{2\pi}\right) \right| = \left| \text{sinc}\left(\frac{2\pi - 0.2}{2\pi}\right) \right| = \left| \text{sinc}\left(1 - \frac{0.1}{\pi}\right) \right|$$

Since $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$, and $x \approx 1 - 0.0318 = 0.968$:

$$\sin(\pi \cdot 0.968) \approx \sin(3.041) \approx 0.0997, \quad \pi(0.968) \approx 3.041$$

$$\text{sinc}(0.968) \approx \frac{0.0997}{3.041} \approx 0.0328$$

Gain reduction ≈ 0.033 (about a 97% drop)

8. Laplace transform of a three-term signal

Recognize: any signal with a scaled/shifted delta + a shifted power-of- t + an exponential-times-power-of- t is always solved the same way: split into three terms (linearity), handle each with exactly one property from the table (delta-scaling, time-shift, frequency-shift), then add.

Term 1 — $3\delta(2t - 6)$. Factor: $2t - 6 = 2(t - 3)$, so $\delta(2t - 6) = \frac{1}{2}\delta(t - 3)$.

$$\mathcal{L}\{3\delta(2t - 6)\} = 3 \cdot \frac{1}{2}e^{-3s} = \frac{3}{2}e^{-3s}$$

Term 2 — $t^2u(t - 2)$. Rewrite $t = (t - 2) + 2$, so $t^2 = (t - 2)^2 + 4(t - 2) + 4$. Distribute across $u(t - 2)$:

$$t^2u(t - 2) = (t - 2)^2u(t - 2) + 4(t - 2)u(t - 2) + 4u(t - 2)$$

Apply time-shift to each (base pairs: $t^2u(t) \rightarrow 2/s^3$, $tu(t) \rightarrow 1/s^2$, $u(t) \rightarrow 1/s$):

$$\mathcal{L}\{t^2u(t - 2)\} = e^{-2s} \left(\frac{2}{s^3} + \frac{4}{s^2} + \frac{4}{s} \right)$$

Term 3 — $te^{2-3t}u(t)$. Split exponent: $e^{2-3t} = e^2e^{-3t}$. Base pair $tu(t) \rightarrow 1/s^2$, frequency-shift by $a = 3$: $s \rightarrow s + 3$.

$$\mathcal{L}\{te^{2-3t}u(t)\} = e^2 \cdot \frac{1}{(s + 3)^2}$$

Add all three:

$$X(s) = \frac{3}{2}e^{-3s} + e^{-2s} \left(\frac{2}{s^3} + \frac{4}{s^2} + \frac{4}{s} \right) + \frac{e^2}{(s+3)^2}$$

9. Prove the time-integral Laplace property

Recognize: "prove a time-domain integral becomes $1/s$ times the transform" is always solved by recognizing the integral as a **convolution with $u(t)$** , then applying the convolution property (multiplication in s).

Step 1: Let $y(t) = \int_{0^-}^t x(\tau) d\tau$. Note $x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau) u(t - \tau) d\tau$; since $u(t - \tau) = 1$ for $\tau < t$ and $x(\tau) = 0$ for $\tau < 0^-$ (causality), this reduces exactly to $\int_{0^-}^t x(\tau) d\tau = y(t)$.

Step 2: Apply the convolution property (Table of properties, #13): $\mathcal{L}\{x_1(t) * x_2(t)\} = X_1(s)X_2(s)$. With $x_1 = x, x_2 = u$:

$$Y(s) = X(s) \cdot \mathcal{L}\{u(t)\} = X(s) \cdot \frac{1}{s}$$

$$\mathcal{L} \left[\int_{0^-}^t x(\tau) d\tau \right] = \frac{1}{s} X(s)$$

10. Bounded signals and BIBO stability

Recognize: this is a pure-derivation question you should have memorized cold — no computation, just the logical chain: bounded input, take magnitude of the convolution integral, pull the bound outside, and conclude what condition on $h(t)$ keeps the output finite.

Definition: $x(t)$ is bounded if $\exists M_x < \infty$ such that $|x(t)| \leq M_x$ for all t .

Derivation:

$$|y(t)| = \left| \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \right| \leq \int_{-\infty}^{\infty} |h(\tau)| |x(t - \tau)| d\tau \leq M_x \int_{-\infty}^{\infty} |h(\tau)| d\tau$$

For $y(t)$ to be guaranteed bounded for *every* bounded input:

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

Connection to poles (recognize-cue for the bonus part): a causal LTI system's stability can be read directly off $H(s)$'s pole locations — BIBO stability requires every pole of $H(s)$ to lie in the **open left half of the s -plane** (strictly negative real part), because each pole $s = -a + jb$ corresponds to a time-domain term like $e^{-at} \cos(bt)u(t)$, whose absolute integral is finite only when $a > 0$.

How to use this practice final

- Work all 10 problems **cold**, timed (aim for ~50–60 min total, matching your midterm pacing).
- For anything you get wrong or can't start, come back and ask me to re-derive that one problem fully, step by step, the way we did for your midterms.
- Notice the recurring "recognize" cues across problems — that vague professor wording almost always maps to one of a small number of named tools: sifting, cascade/associativity, orthogonality, modulation, replica-and-filter, ZOH sinc, table properties, or the bounded-input derivation. Drilling the *recognition* is more valuable than memorizing any one answer, since his wording changes but the underlying pattern doesn't.